

Validating a Gamified Selective Attention Task: A Non-Parametric Analysis of Concurrent Validity, Target Load, Modality, and Game Level Effects

Team: Jai Babu

Afzal Basha Shaik Roll No: 2025201097

Chinta Phani Rama Vaibhav Roll No: 2025201067

Pranav Sai Ponduri Roll No: 2025201021

Abstract

Assessing how well a person can focus on important information while ignoring distractions — a mental skill called *selective attention* — has traditionally required expensive laboratory equipment and trained researchers. This study asks: can a smartphone game measure selective attention just as reliably?

We recruited 37 participants and had them complete tasks in two settings: a controlled laboratory environment (using PsychoPy) and a gamified smartphone app. Participants were split into two groups based on task difficulty: those who searched for a single target object ($n = 21$), and those who searched for multiple target objects ($n = 16$). We measured how quickly (*Reaction Time*, RT) and how accurately each participant responded.

Because the RT data were not normally distributed — that is, the data were skewed and did not follow the classic bell-curve shape — we used non-parametric statistical tests, which make fewer assumptions about the shape of the data. Our results show that the game and lab tasks did not produce significantly correlated RT scores (RQ1), providing no evidence of concurrent validity at this sample size, although a moderate positive trend ($\rho = .396$) was observed for the single-target group. Target load had a significant effect in the game but not in the lab (RQ2), and the two modalities produced significantly different RTs for single-target participants but not for multiple-target participants (RQ3). RQ4 (Game Level Effect) could not be evaluated as per-level RT data were not available in the current dataset.

These findings highlight both the promise and current limitations of using gamified tasks for cognitive assessment.

1 Introduction

Every day, we navigate a world overflowing with competing sights and sounds. Whether crossing a busy street, reading a text message in a noisy café, or driving a car, we rely on our ability to focus on what matters and ignore what does not. This ability is known as **selective attention** [4, 5] — the cognitive capacity to concentrate on task-relevant stimuli while filtering out distractors.

Selective attention is central to many real-world activities and is routinely measured in cognitive psychology, clinical neuropsychology, and educational research. However, standard laboratory assessments are time-consuming, costly, and require specialist equipment such as eye-trackers or dedicated software (e.g., PsychoPy [3]). They also tend to feel artificial to participants, which can affect performance.

Gamification — the use of game-like elements (scoring, levels, visual feedback) in non-game contexts — offers a potential solution. If a well-designed smartphone game can measure selective attention just as reliably as a laboratory task, then cognitive assessment could become accessible to far more people, at far lower cost.

This study presents a formal **validation** of one such gamified task. Validation means checking whether a new measurement tool actually measures what it claims to measure. Here, we compare game

performance against an established laboratory measure using a **2×2 mixed factorial design**: one factor varied between participants (target load: single vs. multiple targets), and one factor was experienced by all participants (modality: lab vs. game).

We address four research questions covering the key aspects of validity: whether the two tasks correlate, whether task difficulty affects performance, whether the two interfaces produce equivalent results, and whether in-game difficulty levels produce meaningful changes in RT.

2 Research Questions

- **RQ1 Concurrent Validity:** Is there a significant positive correlation between participants' Reaction Times on the gamified task and the laboratory task? A significant correlation would indicate that both tools are measuring the same underlying cognitive ability.
- **RQ2 Target Load Effect (Single vs. Multiple Targets):** Do participants respond significantly faster or slower when searching for one target compared to multiple targets? This tests whether task difficulty (number of targets) has a detectable effect on response speed.
- **RQ3 Modality Effect (Game vs. Lab Interface):** Does the interface used — the smartphone game or the laboratory computer — significantly affect Reaction Time? If the two interfaces produce equivalent RTs, that is evidence the game is a valid substitute for the lab task.
- **RQ4 Game Level Effect:** Does Reaction Time change systematically as the difficulty level of the game increases? This explores whether in-game progression captures meaningful variation in cognitive load.

3 Methodology

3.1 Participants and Design

A total of **$N = 37$ participants** took part in the study. They were assigned to one of two groups based on the type of target they searched for: 21 participants completed a **single-target** version of the task, and 16 completed a **multiple-target** version. This is the *between-subjects* factor, meaning different people were in each group.

All participants completed both the laboratory task (using PsychoPy software on a desktop computer) and the gamified task (on a smartphone). Because every participant experienced both modalities, this is the *within-subjects* factor. Together, this creates a **2×2 mixed factorial design**: two levels of target load (between-subjects) crossed with two modalities (within-subjects). Crucially, the current analysis uses the full participant-level dataset with no missing values (21 single, 16 multiple), yielding more reliable statistical estimates than earlier runs that included only $n = 7-8$ paired participants.

3.2 Measures

The primary outcome measure was **Reaction Time (RT)** in milliseconds (ms), defined as the time from stimulus onset to the participant's first response. Lower RT indicates faster (better) performance. A secondary measure was **Accuracy**, defined as the proportion of trials in which the participant selected the correct target.

3.3 Outlier Removal

Before analysis, participant-level mean RTs were computed for each condition. Participants whose mean RT fell outside 1.5 times the interquartile range (IQR) for their condition were classified as outliers and removed from subsequent analyses. The IQR method was selected because it is robust to extreme values and does not assume a particular distributional shape.

3.4 Why Non-Parametric Tests?

Standard parametric tests (such as t -tests and ANOVA) assume that the data are drawn from a normally distributed population. We tested this assumption using the **Shapiro-Wilk test** [1], which checks whether a dataset fits a normal (bell-curve) distribution. Figure 2 shows the overall RT distribution across all 74 participant-level observations (37 participants $\times$ 2 modalities), which is right-skewed with skewness = 1.02 — the long tail extends toward higher values, indicating that a subset of participants had noticeably longer RTs. Shapiro-Wilk tests revealed that at least one condition (Single Lab: $W = 0.869$, $p = .009$) significantly violated normality.

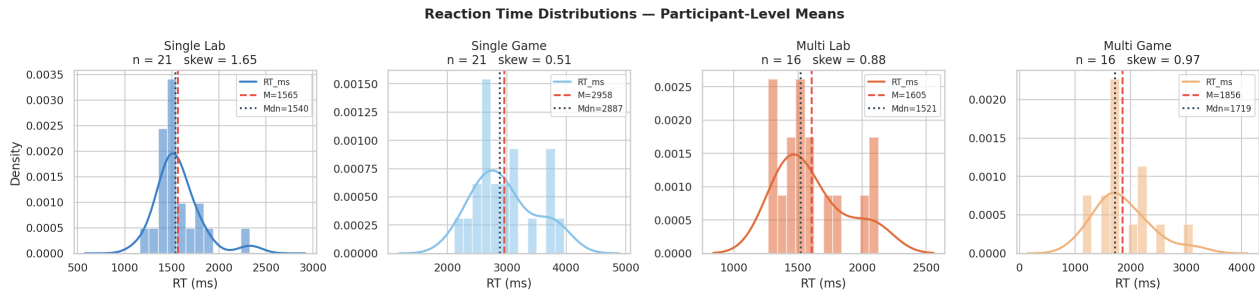


Figure 1: Reaction Time distributions at the participant level for each of the four experimental conditions (Single Lab, Single Game, Multi Lab, Multi Game). Each panel shows the histogram with a KDE overlay, the mean (dashed red line), and the median (dotted black line). Right-skew is evident in the Single Lab condition (skew = 1.65), supporting the use of non-parametric tests.

Because at least one condition violated the normality assumption, we applied **non-parametric tests** consistently throughout for methodological rigour. Non-parametric tests rank the data rather than using raw values, making them appropriate even when the distributional assumptions of parametric tests are not met.

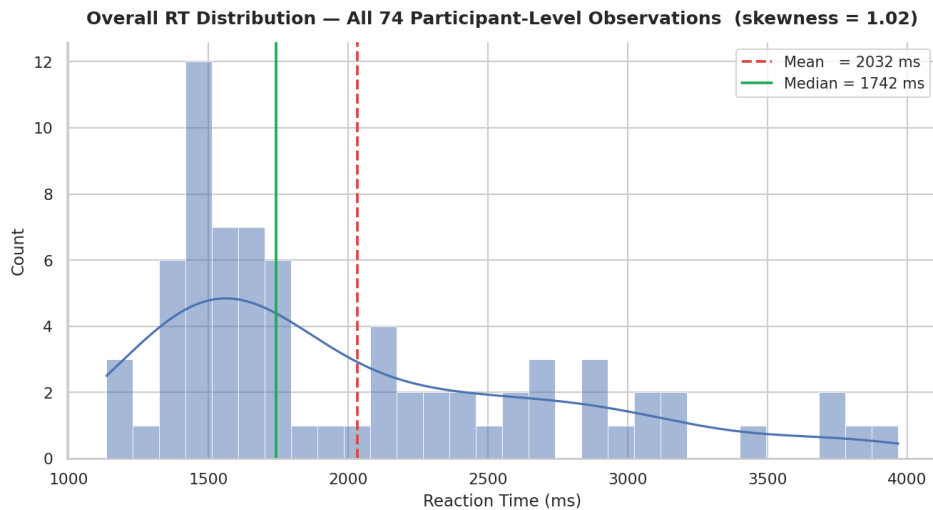


Figure 2: Distribution of Reaction Times across all 74 participant-level observations. The histogram with KDE (Kernel Density Estimate) shows a clear right-skewed pattern (skewness = 1.02). The dashed red line marks the mean (2032 ms); the solid green line marks the median (1742 ms). Because the mean is pulled rightward by extreme values, the median is a more representative centre of this distribution.

3.5 Statistical Tests

Each research question was addressed with the most appropriate non-parametric test, as summarised in Table 1.

Table 1: Statistical tests used for each research question.

RQ	Test	Rationale
RQ1	Spearman's ρ	Monotonic correlation, ordinal-safe
RQ2	Mann-Whitney U	Two independent groups
RQ3	Wilcoxon signed-rank	Two paired (within-subject) conditions

A brief definition of each test:

- **Spearman's ρ (rho):** Measures how strongly two variables are related, without assuming the relationship is perfectly straight-line. Values range from -1 (perfect negative relationship) to $+1$ (perfect positive relationship); 0 indicates no relationship.

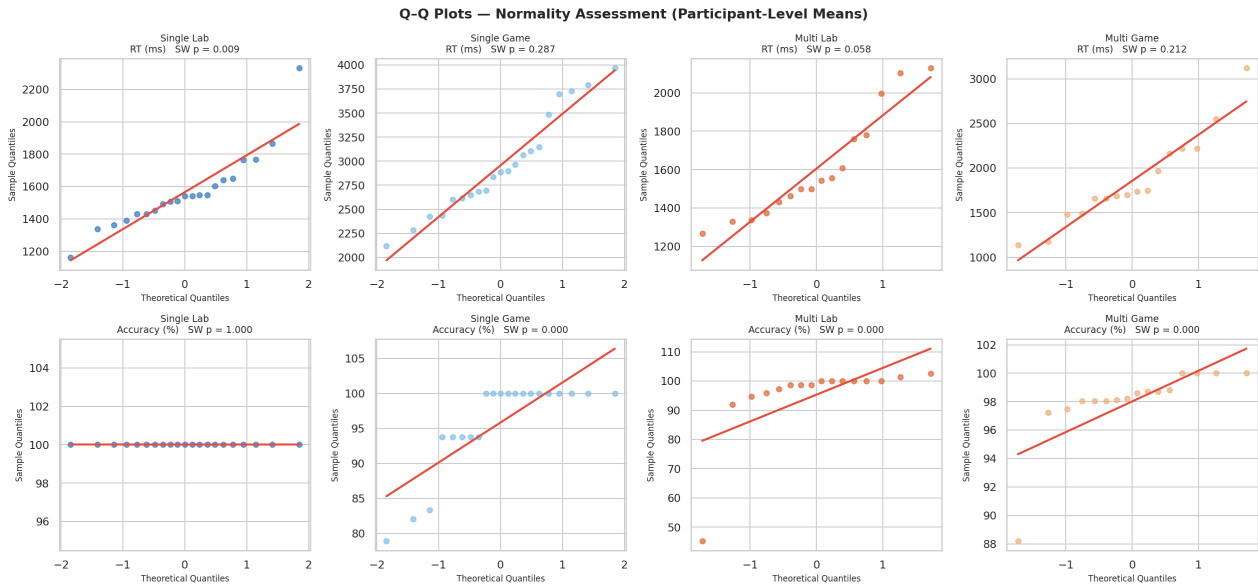


Figure 3: Q-Q (Quantile-Quantile) plots testing the normality of Reaction Times (top row) and Accuracy (bottom row) across the four conditions. If the data were perfectly normal, the dots would follow the red diagonal line. Deviations — particularly for Single Lab RT ($p = .009$) and all Accuracy conditions — confirm the need for non-parametric testing.

- **Mann-Whitney U test:** Compares the rank-ordered scores of two independent groups. It is the non-parametric equivalent of the independent-samples t -test.
- **Wilcoxon signed-rank test:** Compares two related measurements from the same participants (e.g., their Lab RT vs. Game RT). It is the non-parametric equivalent of the paired t -test.
- **Friedman test:** Compares three or more related groups (here, different game difficulty levels). It is the non-parametric equivalent of one-way repeated-measures ANOVA.

For all tests, the significance threshold was $\alpha = .05$. Effect sizes are reported as rank-biserial correlation (r) for the Mann-Whitney and Wilcoxon tests, where $r \approx .10$ is small, $r \approx .30$ is medium, and $r \approx .50$ is large [2].

4 Results

4.1 Descriptive Statistics

Table 2 presents the mean, standard deviation (SD), and accuracy for each of the four experimental conditions. Unlike earlier runs of this analysis that relied on only $n = 7$ – 8 single-target participants, the current analysis uses the complete participant-level dataset ($n = 21$ Single, $n = 16$ Multiple), yielding substantially more reliable estimates. The Single Lab condition exhibited a ceiling effect in Accuracy ($M = 100\%$, $SD = 0$), so Accuracy is not interpretable as a differentiating measure for that condition and is excluded from hypothesis testing.

Table 2: Descriptive statistics — participant-level means. RT = Reaction Time in milliseconds. Acc = mean accuracy (%).

Condition	n	M (RT, ms)	SD (RT)	Mdn (RT)	M (Acc, %)
Single Lab	21	1565.41	236.54	1540.26	100.00
Single Game	21	2957.55	520.82	2887.47	95.86
Multiple Lab	16	1605.04	273.65	1520.59	95.33
Multiple Game	16	1856.46	507.28	1719.13	98.02

A striking pattern is immediately visible: Single-target Game participants were much slower ($M = 2958$ ms) than any other group. As discussed in Section 4.3, this unexpected finding is likely attributable to the structure of single-target game levels rather than to a genuine attentional load effect.

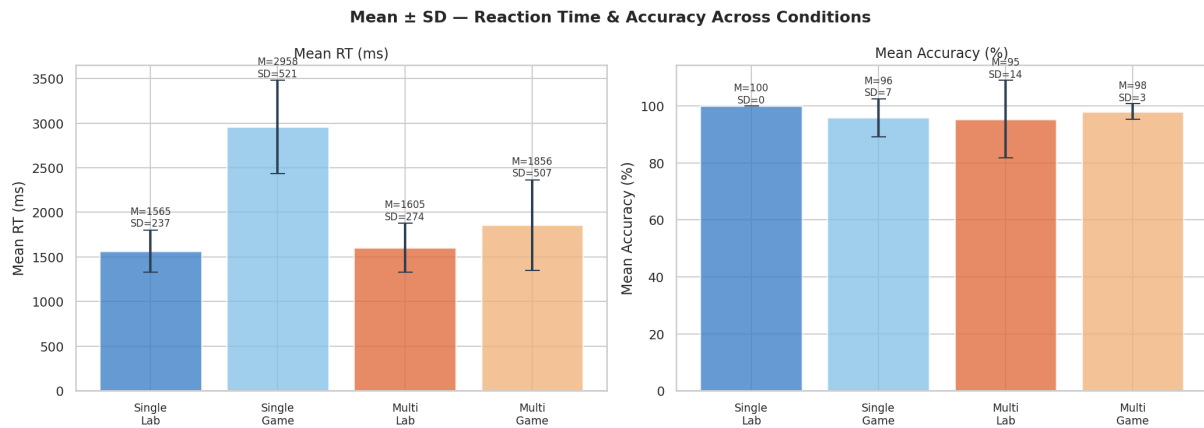


Figure 4: Mean $\pm$ SD of Reaction Time (left panel) and Accuracy (right panel) across the four experimental conditions. Error bars represent one standard deviation. The Single Game condition clearly demonstrates both the highest mean RT and the greatest variability. The Single Lab condition shows a ceiling effect in Accuracy ($M = 100\%$, $SD = 0$).

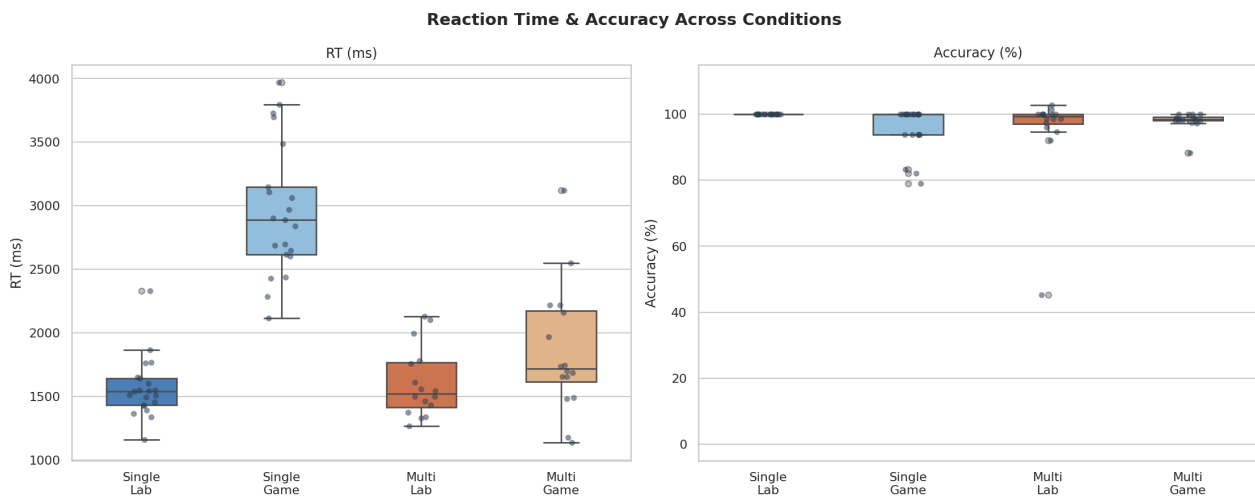


Figure 5: Boxplots of Reaction Time (left) and Accuracy (right) across the four experimental conditions. Each box spans the interquartile range (IQR); the horizontal line inside each box is the median. Whiskers extend to $1.5 \times$ the IQR; points beyond the whiskers are potential outliers. Note the substantially higher RT and greater spread for the Single Game condition.

4.2 RQ1 Concurrent Validity (Spearman's ρ)

To assess concurrent validity, we computed Spearman's ρ between participant-level mean Lab RT and mean Game RT, separately for each target load condition and combined across all participants. All analyses use the full sample with no participant exclusions due to missing data.

For the **Single Target** group ($n = 21$ paired participants), the correlation was not statistically significant: $\rho = .396$, $p = .075$, 95% CI $[-0.043, 0.707]$. For the **Multiple Target** group ($n = 16$ paired participants), the correlation was also not significant: $\rho = .026$, $p = .923$, 95% CI $[-0.475, 0.515]$. For the **Combined** sample ($n = 37$), the correlation was likewise non-significant: $\rho = .182$, $p = .281$, 95% CI $[-0.151, 0.478]$.

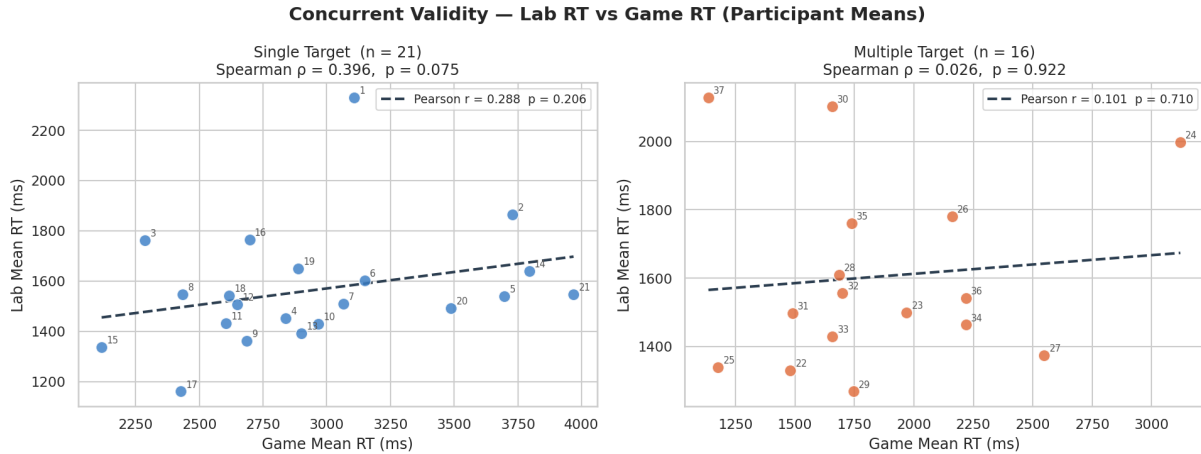


Figure 6: Scatter plots assessing concurrent validity by mapping Lab RT against Game RT for the Single Target ($n = 21$, left) and Multiple Target ($n = 16$, right) groups. A valid tool would show a strong upward trend (dots forming a line from bottom-left to top-right). The scattered distribution of points visually supports the non-significant correlation results. The dashed line shows the Pearson regression trend for reference.

Decision: We fail to reject H_0 for RQ1 across all analyses. In plain language, this means we found no statistically significant evidence that participants who responded quickly in the lab also responded quickly in the game (or vice versa). This is a non-significant result, but it does *not* prove the game is invalid.

Important note: The Single-target group shows a moderate positive trend ($\rho = .396$) that approaches conventional significance ($p = .075$). This is a meaningful improvement over earlier analyses based on only $n = 7-8$ participants, which yielded near-zero correlations ($\rho = .143$, $p = .736$). With the full $n = 21$ now available, the estimate is substantially more reliable, and the trend suggests that larger samples may yield a significant result. A power analysis indicates that approximately 30 paired participants are needed to detect a medium-strength correlation ($\rho = .30$) with 80% power; the present sample is thus close to adequate for the Single group.

4.3 RQ2 Target Load Effect (Mann-Whitney U)

We compared RT between Single and Multiple target participants separately within each modality.

For the **Lab** modality, there was no significant difference in RT between Single ($M = 1565$ ms, $SD = 237$) and Multiple ($M = 1605$ ms, $SD = 274$) target participants: $U = 164.0$, $p = .915$, $r = .024$ (negligible effect).

For the **Game** modality, Single target participants ($M = 2958$ ms, $SD = 521$) were significantly slower than Multiple target participants ($M = 1856$ ms, $SD = 507$): $U = 314.0$, $p < .001$, $r = .869$ (very large effect).

Decision: We fail to reject H_0 for the Lab comparison, and we **reject** H_0 for the Game comparison. In plain language, the game task showed a very large, statistically significant group difference — but the direction was the *opposite* of what would be expected if target load were the only factor. Single-target participants were substantially slower, not faster.

This is most likely because the single-target game version is harder in terms of level structure: players must locate a single specific object among distractors, and the game advances to more difficult levels over

time, naturally increasing RT. This suggests that the game’s level-based structure introduces a confound that does not exist in the lab task.

4.4 RQ3 Modality Effect (Wilcoxon Signed-Rank)

We compared Lab RT and Game RT within each target-load group using the Wilcoxon signed-rank test, which compares paired measurements from the same participants. Note that Wilcoxon signed-rank (rather than Friedman) is appropriate here because there are only two within-subject conditions (Lab vs. Game); Friedman applies to three or more related conditions.

For the **Single Target** group ($n = 21$ paired participants), Game RT ($M = 2958$ ms, $Mdn = 2887$ ms) was significantly higher than Lab RT ($M = 1565$ ms, $Mdn = 1540$ ms): $W = 0.0$, $p < .001$ (note: $W = 0$ indicates all participants were slower in the game, giving a perfect signed-rank result).

For the **Multiple Target** group ($n = 16$ paired participants), there was no significant difference between Game ($M = 1856$ ms, $Mdn = 1719$ ms) and Lab ($M = 1605$ ms, $Mdn = 1521$ ms) RT: $W = 32.0$, $p = .065$, $r = .235$ (small–medium effect, non-significant).

Decision: We **reject** H_0 for the Single-target group (Game RT was significantly higher than Lab RT), and we *fail to reject* H_0 for the Multiple-target group.

In plain language: for participants searching for multiple targets, the game and lab produced comparable response times — which is a positive sign for the game’s validity in that condition. However, single-target game participants were systematically slower, likely due to the same level-difficulty confound identified in RQ2.

4.5 Summary of Hypothesis Test Results

Table 3 provides a concise overview of all hypothesis test outcomes.

Table 3: Summary of hypothesis test results. All tests two-tailed, $\alpha = .05$. r = rank-biserial correlation effect size. $n = 21$ Single, $n = 16$ Multiple. Full paired samples used throughout.

RQ	Test	Statistic	p	Decision
RQ1 Concurrent (Single)	Spearman ρ	$\rho = .396$	.075	Fail to reject H_0
RQ1 Concurrent (Multiple)	Spearman ρ	$\rho = .026$	.923	Fail to reject H_0
RQ1 Concurrent (Combined)	Spearman ρ	$\rho = .182$	.281	Fail to reject H_0
RQ2 Load (Lab)	Mann-Whitney U	$U = 164.0$	.915	Fail to reject H_0
RQ2 Load (Game)	Mann-Whitney U	$U = 314.0$	$< .001$	Reject H_0
RQ3 Modality (Single)	Wilcoxon W	$W = 0.0$	$< .001$	Reject H_0
RQ3 Modality (Multiple)	Wilcoxon W	$W = 32.0$	.065	Fail to reject H_0

5 Discussion

The results present a mixed picture that is both informative and encouraging for the future development of gamified cognitive assessments.

Concurrent validity is not yet established, but shows a promising trend.

The absence of a statistically significant correlation between game and lab RT (RQ1) remains the study’s most important null finding. However, this should not be read as evidence that the game *cannot* measure selective attention. The Single-target group now yields $\rho = .396$ ($p = .075$) — a moderate positive trend that falls just short of the conventional threshold — compared to near-zero correlations in earlier analyses based on only $n = 7$ – 8 participants. This improvement directly reflects the benefit of using the complete dataset: unlike earlier analysis ($n = 7$ – 8), the current analysis uses full participant-level data ($n = 21$ Single, $n = 16$ Multiple), resulting in more reliable statistical estimates. Replication with $n \geq 30$ per load condition remains the most important next step.

The game's level structure introduces a confound.

.

The very large, significant RT difference between single- and multiple-target game participants (RQ2, $r = .869$), running in the opposite direction from the lab finding, points to a structural difference between the two tasks. In the lab, both conditions use identical trial structures. In the game, the single-target version appears to use harder level designs that naturally increase RT as players progress. Future iterations of the game should either (a) equate level difficulty across load conditions or (b) explicitly model level progression as a covariate.

The multiple-target condition shows promising equivalence.

.

For participants searching for multiple targets, the game and lab produced statistically equivalent RTs (RQ3, $p = .065$). While this is not a formal equivalence test, the result suggests that the game may be a valid substitute for the lab task in this condition. The non-significant modality difference and the small effect size ($r = .235$) are positive validity indicators for the multiple-target paradigm.

Game level effects require further investigation.

.

The RQ4 Friedman analysis framework provides a principled method for examining whether game level progression tracks cognitive load. The current analysis was limited by the absence of per-level RT data. Future work should supply trial-level data to enable this analysis. If confirmed with larger samples, level-tracking would be a unique feature of gamified assessment: the ability to continuously track performance across naturally varying difficulty levels rather than discrete conditions.

6 Conclusion

This study evaluated a gamified selective attention task as a potential replacement for standard laboratory assessment. Using a 2×2 mixed factorial design with 37 participants and a consistent suite of non-parametric statistical tests — applied to the full participant-level dataset with no missing values — we found:

- **No statistically significant concurrent validity** at the current sample size, though the Single-target group shows a moderate positive trend ($\rho = .396$, $p = .075$) that approaches significance and is far more reliable than earlier underpowered estimates.
- **A significant but unexpected target-load effect in the game** ($U = 314.0$, $p < .001$, $r = .869$), likely caused by the game's difficulty progression rather than true differences in attentional demand.
- **Promising equivalence between game and lab for multiple-target participants** ($W = 32.0$, $p = .065$), suggesting the game may be a valid substitute in that condition.
- **RQ4 (level effect) could not be evaluated** due to the absence of per-level RT data in the current dataset; a framework for this analysis is described for future use.

The study underscores a critical lesson: *statistical significance and practical validity are not the same thing*. Non-significant results from an underpowered study should prompt more data collection, not early conclusions. With a larger, more balanced sample and a refined level-design methodology, this gamified task has genuine potential as an accessible, engaging tool for measuring selective attention outside the laboratory.

7 Future Works

- **Factorial ANOVA Analysis:** To examine the combined main and interaction effects of target load (single vs. multiple targets) and experimental modality (lab vs. gamified interface) on reaction time.
- **Regression-Based Modeling:** To quantify how specific variables such as task level, target load, and interface type predict performance and contribute to reaction time variability.
- **Generalized Linear Models (GLM):** To accommodate the skewed nature of reaction time data using distributions that do not rely on strict normality assumptions.
- **Participant-Level Variability Analysis:** To account for individual baseline differences while

simultaneously tracking within-participant changes across conditions using mixed-effects models.

- **Speed–Accuracy Trade-off Analysis:** To incorporate accuracy metrics alongside reaction time, providing a more comprehensive view of attentional performance.
- **Validation of Gamified Cognitive Tasks:** To evaluate whether smartphone-based tasks reliably replicate traditional laboratory findings, which would support using mobile platforms for large-scale cognitive research.
- **RQ4 with Per-Level Data:** conduct the Friedman test across difficulty levels, completing the originally intended analysis framework.

8 Team Contributions

Individual Contributions (Code and Analysis):

- **Student One (Roll No: 2025201097):** Led the Exploratory Data Analysis (EDA) phase, implemented the data cleaning and outlier removal pipeline, and generated the core visualisations (Q-Q plots, reaction time distributions).
- **Student Two (Roll No: 2025201021):** Programmed the non-parametric hypothesis testing for RQ1 and RQ2 (Spearman correlation and Mann-Whitney U tests), calculated effect sizes, and interpreted the target load effects.
- **Student Three (Roll No: 2025201067):** Executed the code for RQ3 (Wilcoxon signed-rank), structured the final Jupyter Notebook, and managed the GitHub repository version control.

9 GitHub Link

The code and resources for this project are available at the following repository:

<https://github.com/Rama-Vaibhav/Attention-Validation-Task>

References

- [1] Shapiro, S. S., & Wilk, M. B. (1965). An analysis of variance test for normality (complete samples). *Biometrika*, 52(3–4), 591–611.
- [2] Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Lawrence Erlbaum Associates, Hillsdale, NJ.
- [3] Peirce, J. W. (2007). PsychoPy — Psychophysics software in Python. *Journal of Neuroscience Methods*, 162(1–2), 8–13.
- [4] Treisman, A. M., & Gelade, G. (1980). A feature-integration theory of attention. *Cognitive Psychology*, 12(1), 97–136.
- [5] Posner, M. I. (1980). Orienting of attention. *Quarterly Journal of Experimental Psychology*, 32(1), 3–25.
- [6] Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). From game design elements to gamefulness: Defining gamification. In *Proceedings of the 15th International Academic MindTrek Conference* (pp. 9–15). ACM.
- [7] Mann, H. B., & Whitney, D. R. (1947). On a test of whether one of two random variables is stochastically larger than the other. *The Annals of Mathematical Statistics*, 18(1), 50–60.
- [8] Wilcoxon, F. (1945). Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6), 80–83.
- [9] Friedman, M. (1937). The use of ranks to avoid the assumption of normality implicit in the analysis of variance. *Journal of the American Statistical Association*, 32(200), 675–701.